# PRIMES is in P: A Complete Guided Exposition

Based on Agrawal, Kayal, Saxena (2002)

Study Notes

## Contents


**1 What the paper proves and why it matters** **3**
- 1.1 The problem . . . 3
- 1.2 Complexity landscape before AKS . . . 3

**2 The key identity (Lemma 2.1)** **4**
- 2.1 Why this identity works . . . 4
- 2.2 Why we cannot use this identity directly . . . 5

**3 The key trick: working mod $X^r - 1$** **5**
- 3.1 What "mod $X^r - 1$" means . . . 5
- 3.2 The false positive problem and the fix . . . 5
- 3.3 How to choose $r$ . . . 6

**4 The algorithm** **7**
- 4.1 What each step does and why . . . 7

**5 Correctness proof** **7**
- 5.1 Setup . . . 7
- 5.2 Extracting polynomial identities . . . 8
- 5.3 Introspective numbers . . . 8
- 5.4 Two closure lemmas . . . 9
- 5.5 Building the sets $I$ and $P$ . . . 9
- 5.6 Group $G$: residues of $I$ modulo $r$ . . . 10
- 5.7 Group $\hat{G}$: images of $P$ in a finite field . . . 10
- 5.8 Lower bound on $|\hat{G}|$ — Lemma 4.7 (Lenstra) . . . 11
- 5.9 Upper bound on $|\hat{G}|$ — Lemma 4.8 . . . 11
- 5.10 The squeeze and conclusion . . . 12

**6 Time complexity** **12**

6.1 Improvements . . . . . 13

**7 Open questions** **13**

**8 The complete proof in one place** **13**

# 1 What the paper proves and why it matters

## 1.1 The problem

Given an integer $n > 1$, is $n$ prime?

The ancient Greeks knew a method: try dividing $n$ by every integer up to $\sqrt{n}$. If any divides evenly, $n$ is composite; otherwise $n$ is prime. The trouble is efficiency. The *size* of the input $n$—the number of digits needed to write it—is about $\log n$. Trial division takes $\Omega(\sqrt{n})$ steps, which is *exponential* in the input size. For a 300-digit number this is around $10^{150}$ steps: completely infeasible.

**What "efficient" means**

An algorithm is *efficient* (runs in **polynomial time**) if its number of steps is bounded by a polynomial in the input size. For primality testing, the input has $\lfloor \log_2 n \rfloor + 1$ bits, so efficient means the algorithm terminates in at most $(\log n)^k$ steps for some fixed constant $k$, no matter how large $n$ is.

## 1.2 Complexity landscape before AKS

It was long known that PRIMES $\in$ NP $\cap$ co-NP:

- **co-NP**: a non-trivial factor of $n$ is a short, efficiently verifiable certificate of compositeness.
- **NP**: Pratt (1975) showed every prime has a short certificate (a primitive root and the factorisation of $p - 1$), verifiable in polynomial time.

Membership in NP $\cap$ co-NP strongly suggested PRIMES $\in$ P but did not prove it. Partial results:

- **Miller (1976):** Deterministic poly-time, assuming the Extended Riemann Hypothesis (ERH).
- **Rabin (1980), Solovay–Strassen (1977):** Unconditional but randomised (Monte Carlo).
- **Adleman–Pomerance–Rumely (1983):** Deterministic, time $(\log n)^{O(\log\log\log n)}$—almost polynomial but not quite.
- **Goldwasser–Kilian (1986), Adleman–Huang (1992):** Randomised elliptic curve methods, expected polynomial time.

**The AKS result**

There exists a **deterministic, unconditional, polynomial-time** algorithm that correctly decides whether any integer $n > 1$ is prime. The running time is $\widetilde{O}(\log^{21/2} n)$, and heuristically $\widetilde{O}(\log^6 n)$ under a conjecture on Sophie Germain primes. This places PRIMES firmly in **P**.

# 2 The key identity (Lemma 2.1)

The entire algorithm rests on one algebraic fact.

**Lemma 2.1**

Let $a \in \mathbb{Z}$, $n \geq 2$, $\gcd(a, n) = 1$. Then

$$n \text{ is prime} \iff (X + a)^n \equiv X^n + a \pmod{n}$$

as polynomials over $\mathbb{Z}_n$.

## 2.1 Why this identity works

Expand $(X + a)^n$ by the binomial theorem:

$$(X + a)^n = \sum_{i=0}^{n} \binom{n}{i} a^{n-i} X^i = X^n + \binom{n}{1} a^{n-1} X^{n-1} + \cdots + \binom{n}{n-1} aX + a^n.$$

We want this to equal $X^n + a$ modulo $n$. The endpoint terms give $X^n$ and $a^n$. Fermat's Little Theorem gives $a^n \equiv a \pmod{p}$ when $p$ is prime. So everything reduces to: do all *middle coefficients* $\binom{n}{i}$ vanish modulo $n$ for $0 < i < n$?

**Case 1: $n$ is prime.** For $0 < i < n$:

$$\binom{n}{i} = \frac{n!}{i!\,(n-i)!}.$$

The numerator contains $n$ as a factor. Since $n$ is prime and $i, n - i < n$, neither $i!$ nor $(n - i)!$ contains $n$ as a factor. So $n \mid \binom{n}{i}$ for all $0 < i < n$, and every middle coefficient vanishes. ✓

**Case 2: $n$ is composite.** Let $q$ be a prime factor with $q^k \| n$ (meaning $q^k \mid n$ but $q^{k+1} \nmid n$). Consider $\binom{n}{q}$:

$$\binom{n}{q} = \frac{n(n-1)\cdots(n-q+1)}{q!}.$$

The factor $q^k$ appears in the numerator from $n$. The denominator $q!$ contributes exactly $q^1$. So $q^{k-1} \| \binom{n}{q}$. Since $q^k \mid n$, the coefficient of $X^q$ is not zero mod $n$. ✗

**Concrete examples**

$n = 5$ **(prime),** $a = 1$**:** $(X+1)^5 = X^5 + 5X^4 + 10X^3 + 10X^2 + 5X + 1$. Mod 5: $5 \to 0$, $10 \to 0$, so $(X+1)^5 \equiv X^5 + 1 \pmod 5$. Since $X^5 + 1 = X^n + a$: ✓

$n = 4$ **(composite),** $a = 1$**:** $(X+1)^4 = X^4 + 4X^3 + 6X^2 + 4X + 1$. Mod 4: $(X+1)^4 \equiv X^4 + 2X^2 + 1$. But $X^4 + 1 \neq X^4 + 2X^2 + 1$. The $X^2$ coefficient is 2, not 0. ✗

### 2.2 Why we cannot use this identity directly

The identity perfectly characterises primes. The problem is cost: evaluating $(X+a)^n$ as a polynomial requires computing $n+1$ coefficients, taking $\Omega(n)$ time—exponential in $\log n$. We need a way to compress the computation.

## 3 The key trick: working mod $X^r - 1$

### 3.1 What "mod $X^r - 1$" means

We work in the ring $\mathbb{Z}_n[X]/(X^r - 1)$: polynomial arithmetic modulo both $n$ (the integer) and $X^r - 1$ (the polynomial).

Setting $X^r - 1 = 0$ means $X^r \equiv 1$. So exponents on $X$ wrap around modulo $r$, like a clock with $r$ positions:

$$X^r \equiv 1, \quad X^{r+1} \equiv X, \quad X^{r+2} \equiv X^2, \quad \ldots$$

A polynomial with $n+1$ terms collapses to at most $r$ terms, since all coefficients of $X^0, X^r, X^{2r}, \ldots$ get summed into one slot, and similarly for each residue class mod $r$.

**Clock example with $r = 4$**

$X^4 \equiv 1$, $X^5 \equiv X$, $X^6 \equiv X^2$, $X^7 \equiv X^3$, $X^8 \equiv 1$, and so on.
The polynomial $2X^7 + X^6 + 3X^5 + X^4 + X^2 + 1$ reduces modulo $X^4 - 1$ to: $2X^3 + X^2 + 3X + 1 + X^2 + 1 = 2X^3 + 2X^2 + 3X + 2$. Only 4 terms remain instead of 8.

Instead of checking $(X+a)^n \equiv X^n + a \pmod n$ exactly, we check the cheaper condition:

$$(X+a)^n \equiv X^n + a \pmod{X^r - 1,\ n}. \tag{$*$}$$

Computing the left side via repeated squaring requires $O(\log n)$ multiplications of degree-$r$ polynomials with $O(\log n)$-bit coefficients, costing $\widetilde{O}(r \log^2 n)$ total—polynomial in $\log n$ when $r = \mathrm{poly}(\log n)$.

### 3.2 The false positive problem and the fix

Primes still satisfy $(*)$: the original identity holds for primes, so reducing mod $X^r - 1$ only preserves it. However, some composites can now *also* satisfy $(*)$ for certain choices of $a$ and $r$. We have lost information by collapsing the ring.

**Verifying $(*)$ for a prime and a composite**

$n = 5$ **(prime),** $r = 3$**,** $a = 1$**.**
Left side: $(X+1)^5 = X^5 + 5X^4 + 10X^3 + 10X^2 + 5X + 1$. Mod 5: $\equiv X^5 + 1$. Mod $X^3 - 1$: $X^5 = X^{3+2} \equiv X^2$, so $\equiv X^2 + 1$.
Right side: $X^5 + 1$. Mod $X^3 - 1$: $X^5 \equiv X^2$, so $\equiv X^2 + 1$.
Both sides $= X^2 + 1$. ✓

$n = 4$ **(composite),** $r = 3$**,** $a = 1$**.**
Left: $(X+1)^4 = X^4 + 4X^3 + 6X^2 + 4X + 1$. Mod 4: $\equiv X^4 + 2X^2 + 1$. Mod $X^3 - 1$: $X^4 \equiv X$, so $\equiv X + 2X^2 + 1$.
Right: $X^4 + 1$. Mod $X^3 - 1$: $\equiv X + 1$.
$X + 2X^2 + 1 \neq X + 1$. The $X^2$ coefficient exposes the composite. ✗

**The fix:** test $(*)$ for *many values* of $a$. For an appropriately chosen $r$, if $(*)$ holds for all $a = 1, \ldots, \ell$ where $\ell = \lfloor \sqrt{\varphi(r)} \log n \rfloor$, then $n$ must be a prime power. Since proper prime powers are ruled out by step 1, the algorithm concludes $n$ is prime. Why this works is the content of the correctness proof (Section 5).

## 3.3 How to choose $r$

We need $r$ with $\mathrm{ord}_r(n) > \log^2 n$, where $\mathrm{ord}_r(n)$ is the smallest $k$ with $n^k \equiv 1 \pmod{r}$. This condition ensures a certain group (built in the proof) is large enough.

**Lemma 4.3**

There always exists $r \leq \max\{3, \lceil \log^5 n \rceil\}$ satisfying $\mathrm{ord}_r(n) > \log^2 n$.

*Proof sketch.* The product $n^{\lfloor \log B \rfloor} \cdot \prod_{i=1}^{\lfloor \log^2 n \rfloor} (n^i - 1)$ with $B = \lceil \log^5 n \rceil$ is less than $2^B$. By the lcm lemma ($\mathrm{lcm}(1, \ldots, B) \geq 2^B$), this product cannot be divisible by every integer from 1 to $B$. The smallest $r \leq B$ not dividing the product satisfies $\gcd(r, n) = 1$ and $\mathrm{ord}_r(n) > \log^2 n$. $\square$

Since $r \leq \log^5 n$, the loop bound is $\ell = O(\sqrt{r} \log n) = O(\log^{7/2} n)$—polynomial in $\log n$.

# 4 The algorithm

**The AKS Algorithm**

**Input:** Integer $n > 1$.

**Step 1.** If $n = a^b$ for some $a \in \mathbb{N}$, $b > 1$: output COMPOSITE.

**Step 2.** Find the smallest $r$ with $\mathrm{ord}_r(n) > \log^2 n$.

**Step 3.** If $1 < \gcd(a, n) < n$ for some $a \leq r$: output COMPOSITE.

**Step 4.** If $n \leq r$: output PRIME.

**Step 5.** For $a = 1$ to $\lfloor \sqrt{\varphi(r)} \log n \rfloor$: if $(X + a)^n \not\equiv X^n + a \pmod{X^r - 1, n}$, output COMPOSITE.

**Step 6.** Output PRIME.

## 4.1 What each step does and why

**Step 1: Rule out perfect powers.** If $n = a^b$ with $b > 1$, then $n$ is composite. Check by trying all $b$ from 2 to $\log_2 n$ and testing if $n$ has an integer $b$-th root. Cost: $\widetilde{O}(\log^3 n)$. This step is needed because the correctness proof will conclude $n$ must be a prime power; step 1 pre-emptively eliminates the higher prime powers so we can conclude $n = p$ at the end.

**Step 2: Find $r$.** Sets up the polynomial ring. Search for the smallest $r$ with $\mathrm{ord}_r(n) > \log^2 n$. By Lemma 4.3, $r \leq \log^5 n$ exists. For each candidate $r$, test $n^k \not\equiv 1 \pmod{r}$ for $k \leq \log^2 n$. Cost: $\widetilde{O}(\log^7 n)$.

**Step 3: Trial division up to $r$.** For each $a \leq r$, compute $\gcd(a, n)$. If any gcd is strictly between 1 and $n$, we have a non-trivial factor of $n$. Cost: $O(\log^6 n)$. This also guarantees $\gcd(n, r) = 1$, which is needed by the correctness proof.

**Step 4: Handle tiny $n$.** If $n \leq r$ and $n$ survived steps 1–3, then $n$ is prime. Why? A composite $n$ has a factor $\leq \sqrt{n} \leq r$, which step 3 would have caught. By Lemma 4.3 this only triggers for $n \leq 5{,}690{,}034$.

**Step 5: The main loop.** For each $a = 1, \ldots, \ell$, verify $(X+a)^n \equiv X^n + a \pmod{X^r - 1, n}$. Any failure implies COMPOSITE. Cost per check: $\widetilde{O}(r \log^2 n)$; total: $\widetilde{O}(\log^{21/2} n)$.

**Step 6: Output Prime.** Anything that survives all steps is prime, as proved in Section 5.

# 5 Correctness proof

We prove: *if the algorithm outputs* PRIME *at step 6, then $n$ is prime.*

## 5.1 Setup

Suppose the algorithm reaches step 6. We know:

- $n$ is not a perfect power (step 1 passed).
- $\gcd(a, n) = 1$ for all $a \le r$; in particular $\gcd(n, r) = 1$ (step 3 passed).
- $n > r$ (step 4 did not trigger).
- $\mathrm{ord}_r(n) > \log^2 n$ (by step 2).
- For every $a$, $0 \le a \le \ell$:

$$(X + a)^n \equiv X^n + a \pmod{X^r - 1, n}.\text{(step 5)}$$

Let $p$ be **any prime factor of** $n$. Such a factor exists since $n > 1$. We have $p \mid n$. Also $p > r$: if $p \le r$ then $\gcd(p, n) = p \in (1, n)$ and step 3 would have caught it.

## 5.2 Extracting polynomial identities

**Identity (A): step 5 weakened to mod $p$.** Since $p \mid n$, any congruence mod $n$ is also one mod $p$:

$$(X + a)^n \equiv X^n + a \pmod{X^r - 1,\ p}, \quad 0 \le a \le \ell. \tag{A}$$

**Identity (B): Lemma 2.1 applied to the actual prime $p$.**

$$(X + a)^p \equiv X^p + a \pmod{X^r - 1,\ p}, \quad 0 \le a \le \ell. \tag{B}$$

**Identity (C): $n/p$ behaves like a prime.** Raise both sides of (B) to the power $n/p$:

$$[(X + a)^p]^{n/p} \equiv [X^p + a]^{n/p} \pmod{X^r - 1,\ p}.$$

The left side is $(X + a)^n$. By (A), $(X + a)^n \equiv X^n + a$. Also $X^n = (X^p)^{n/p}$, so:

$$(X^p + a)^{n/p} \equiv (X^p)^{n/p} + a \pmod{X^r - 1,\ p}. \tag{C}$$

Writing $Y = X^p$: $(Y + a)^{n/p} \equiv Y^{n/p} + a$. This says $n/p$ satisfies the same polynomial identity that a prime would.

## 5.3 Introspective numbers

**Definition: introspective**

A number $m \in \mathbb{N}$ is *introspective for* polynomial $f(X)$ if

$$[f(X)]^m \equiv f(X^m) \pmod{X^r - 1,\ p}.$$

**Intuition**

Introspectivity says: "raising $f$ to the power $m$ is the same as substituting $X^m$ into $f$." For a monomial this is trivial: $(X^k)^m = X^{km} = (X^m)^k$. For a sum of monomials it is a non-trivial structural condition. Think of it as saying that $m$ "commutes with" the polynomial $f$ in a specific way.

From our identities:

- $p$ is introspective for $(X + a)$ for each $0 \le a \le \ell$ (identity B).
- $n/p$ is introspective for $(X + a)$ for each $0 \le a \le \ell$ (identity C rewritten).

## 5.4 Two closure lemmas

**Lemma 4.5: products of introspective numbers**

If $m$ and $m'$ are both introspective for $f(X)$, then $m \cdot m'$ is also introspective for $f(X)$.

*Proof.*

$$\begin{aligned}
[f(X)]^{m \cdot m'} &= \left[[f(X)]^m\right]^{m'} \\
&\equiv [f(X^m)]^{m'} && \text{(introspectivity of } m\text{)} \\
&\equiv f((X^m)^{m'}) = f(X^{m \cdot m'}) && \text{(introspectivity of } m'\text{, with } X \mapsto X^m\text{)}
\end{aligned}$$

The last step uses the fact that $X^r - 1$ divides $X^{mr} - 1$, so the congruence mod $X^{mr} - 1$ implies one mod $X^r - 1$. $\square$

**Lemma 4.6: products of polynomials**

If $m$ is introspective for both $f(X)$ and $g(X)$, then $m$ is introspective for $f(X) \cdot g(X)$.

*Proof.*

$$[f(X) \cdot g(X)]^m = [f(X)]^m \cdot [g(X)]^m \equiv f(X^m) \cdot g(X^m) = [f \cdot g](X^m). \quad \square$$

## 5.5 Building the sets $I$ and $P$

Starting from the two seeds, the closure lemmas let us expand to large sets.

**Expanding via Lemma 4.5:** By multiplying $p$ and $n/p$ together in any combination and any number of times, every element of

$$I = \left\{ \left(\tfrac{n}{p}\right)^i \cdot p^j \;\middle|\; i, j \ge 0 \right\}$$

is introspective for each $(X + a)$, $0 \le a \le \ell$. (We start with $i = j = 0$ giving 1, then $i = 1, j = 0$ giving $n/p$, and $i = 0, j = 1$ giving $p$, and build up by Lemma 4.5.)

**Expanding via Lemma 4.6:** Since every $m \in I$ is introspective for each individual $(X + a)$, Lemma 4.6 lets us take products. Every element of

$$P = \left\{ \prod_{a=0}^{\ell} (X + a)^{e_a} \;\middle|\; e_a \geq 0 \right\}$$

is a product of the building-block polynomials $(X + 0), (X + 1), \ldots, (X + \ell)$. Lemma 4.6 applied repeatedly shows every $m \in I$ is introspective for every $f \in P$.

**What $P$ is concretely**

$P$ contains all polynomials you can build by multiplying copies of $(X + 0), (X + 1), \ldots, (X + \ell)$ in any combination with any repetition. Examples: $X$, $(X+1)^3(X+2)^2$, $X(X + 1) \cdots (X + \ell)$, and the constant 1 (empty product). Lemma 4.6 says: if $m$ works introspectively for each building block, it works for every product of building blocks.

## 5.6 Group $G$: residues of $I$ modulo $r$

Since $\gcd(n, r) = \gcd(p, r) = 1$, every element of $I$ is a unit in $\mathbb{Z}_r^*$. The residues

$$G = \{m \bmod r \mid m \in I\} \subseteq \mathbb{Z}_r^*$$

form a subgroup of $\mathbb{Z}_r^*$, generated by $n \bmod r$ and $p \bmod r$. Let $t = |G|$.

**Size of $G$**

$t = |G| > \log^2 n$.

*Why?* $G$ contains all powers of $n$ modulo $r$. The cyclic subgroup generated by $n \bmod r$ alone has order $\mathrm{ord}_r(n) > \log^2 n$ (guaranteed by step 2). So $t \geq \mathrm{ord}_r(n) > \log^2 n$.

## 5.7 Group $\hat{G}$: images of $P$ in a finite field

The $r$-th cyclotomic polynomial $Q_r(X)$ over $\mathbb{F}_p$ divides $X^r - 1$ and factors into irreducible polynomials of degree $\mathrm{ord}_r(p)$ over $\mathbb{F}_p$. Let $h(X)$ be one such irreducible factor. Since $p > r$, we have $\mathrm{ord}_r(p) > 1$, so $\deg h > 1$. Define the field:

$$\mathcal{F} = \mathbb{F}_p[X]/(h(X)).$$

In $\mathcal{F}$, $X$ is a primitive $r$-th root of unity (since $h \mid Q_r \mid X^r - 1$, but $h$ has degree $> 1$ so it is not $(X - 1)$, hence $X \neq 1$ in $\mathcal{F}$, i.e., $X$ has order exactly $r$).

Since $\deg h > 1$, the polynomial $h$ has no linear factors, so $X + a \neq 0$ in $\mathcal{F}$ for any integer $a$. Therefore every element of $P$ maps to a nonzero element of $\mathcal{F}$, and the images form a subgroup:

$$\hat{G} = \{f \bmod h(X) \mid f \in P\} \subseteq \mathcal{F}^*,$$

generated by $X, X + 1, X + 2, \ldots, X + \ell$ in $\mathcal{F}$.

## 5.8 Lower bound on $|\hat{G}|$ — Lemma 4.7 (Lenstra)

**Lemma 4.7**

$|\hat{G}| \geq \binom{t+\ell}{t-1} > n^{\sqrt{t}}$.

*Proof.* We show that distinct polynomials in $P$ of degree $< t$ give *distinct* elements of $\hat{G}$.

Suppose $f, g \in P$, $\deg f, \deg g < t$, and $f \neq g$ as polynomials, but $f \equiv g$ in $\mathcal{F}$. Since every $m \in I$ is introspective for both $f$ and $g$:

$$f(X^m) \equiv [f(X)]^m \equiv [g(X)]^m \equiv g(X^m) \quad \text{in } \mathcal{F}.$$

So $X^m$ is a root of $Q(Y) = f(Y) - g(Y)$ for every $m \in G$. Each such $X^m$ is a primitive $r$-th root of unity in $\mathcal{F}$ (since $\gcd(m, r) = 1$), and different $m \in G$ give different $X^m$ (since $X$ has order $r$). So $Q$ has $t$ distinct roots. But $\deg Q < t$—contradiction. Therefore $f = g$, contradicting our assumption. $\square$

So all polynomials in $P$ of degree $< t$ inject into $\hat{G}$. The number of such polynomials is $\binom{t+\ell}{t-1}$ (choosing exponents $e_0, \ldots, e_\ell \geq 0$ summing to $< t$). Using $t > \log^2 n$ and $\ell \geq \lfloor \sqrt{t} \log n \rfloor$:

$$\binom{t+\ell}{t-1} \geq \binom{2\lfloor \sqrt{t} \log n \rfloor + 1}{\lfloor \sqrt{t} \log n \rfloor} > 2^{\lfloor \sqrt{t} \log n \rfloor + 1} \geq n^{\sqrt{t}}.$$

## 5.9 Upper bound on $|\hat{G}|$ — Lemma 4.8

**Lemma 4.8**

If $n$ is not a power of $p$, then $|\hat{G}| \leq n^{\sqrt{t}}$.

*Proof.* Consider:

$$\hat{I} = \left\{ \left(\tfrac{n}{p}\right)^i \cdot p^j \;\middle|\; 0 \leq i, j \leq \lfloor \sqrt{t} \rfloor \right\} \subseteq I.$$

If $n$ is not a power of $p$, then $n/p$ and $p$ are multiplicatively independent, so $\hat{I}$ has $(\lfloor \sqrt{t} \rfloor + 1)^2 > t$ distinct elements as integers.

Since $G$ has only $t$ residues mod $r$, by pigeonhole two elements of $\hat{I}$ agree mod $r$: say $m_1 > m_2$ with $m_1 \equiv m_2 \pmod{r}$. Then $X^{m_1} \equiv X^{m_2}$ in $\mathcal{F}$. For any $f \in \hat{G}$:

$$f^{m_1} \equiv f(X^{m_1}) \equiv f(X^{m_2}) \equiv f^{m_2} \quad \text{in } \mathcal{F}.$$

Every element of $\hat{G}$ is a root of $Q'(Y) = Y^{m_1} - Y^{m_2}$, which has degree

$$m_1 \leq \left(\tfrac{n}{p} \cdot p\right)^{\lfloor \sqrt{t} \rfloor} = n^{\lfloor \sqrt{t} \rfloor} \leq n^{\sqrt{t}}.$$

So $|\hat{G}| \leq n^{\sqrt{t}}$. $\square$

### 5.10 The squeeze and conclusion

We have both:

$$|\hat{G}| > n^{\sqrt{t}} \quad \text{(Lemma 4.7, always)}$$
$$|\hat{G}| \leq n^{\sqrt{t}} \quad \text{(Lemma 4.8, if } n \text{ is not a power of } p\text{)}$$

These are contradictory. So $n$ *is* a power of $p$: $n = p^k$ for some $k \geq 1$.

Step 1 ruled out $k > 1$. Therefore $k = 1$, i.e., $n = p$—prime. □

**Theorem 4.1**

The AKS algorithm outputs PRIME if and only if $n$ is prime.

**The big picture of the proof**

The proof is a squeeze on a group $\hat{G}$. On one side: the polynomial identity passing for many $a$'s forces $\hat{G}$ to be *large*—distinct polynomials must map to distinct group elements, because introspectivity translates group elements into roots of a difference polynomial, and a nonzero polynomial of degree $< t$ cannot have $t$ roots. On the other side: if $n$ were not a prime power, a pigeonhole argument forces $\hat{G}$ to be *small*—every element of $\hat{G}$ would satisfy a low-degree equation. Both cannot hold simultaneously, so $n$ is a prime power, and step 1 ensures it is actually prime.

## 6 Time complexity

**Theorem 5.1**

The asymptotic time complexity of the AKS algorithm is $\widetilde{O}(\log^{21/2} n)$.

*Proof.* We use the fact that arithmetic on $m$-bit integers and degree-$d$ polynomials with $m$-bit coefficients costs $\widetilde{O}(m)$ and $\widetilde{O}(d \cdot m)$ steps respectively.

- **Step 1:** $\widetilde{O}(\log^3 n)$.
- **Step 2:** $O(\log^5 n)$ candidate values of $r$; for each, $O(\log^2 n)$ multiplications mod $r$ at cost $\widetilde{O}(\log r)$ each. Total: $\widetilde{O}(\log^7 n)$.
- **Step 3:** $O(r) = O(\log^5 n)$ gcd computations at $O(\log n)$ each. Total: $O(\log^6 n)$.
- **Step 4:** $O(\log n)$.
- **Step 5:** $\ell = O(\sqrt{r} \log n) = O(\log^{7/2} n)$ iterations; each does $O(\log n)$ polynomial multiplications at cost $\widetilde{O}(r \log n)$ each. Per iteration: $\widetilde{O}(r \log^2 n)$. Total: $\widetilde{O}(\log^{7/2} n \cdot \log^5 n \cdot \log^2 n) = \widetilde{O}(\log^{21/2} n)$.

Step 5 dominates. □

### 6.1 Improvements

**To $\widetilde{O}(\log^{15/2} n)$ (unconditional).** Fouvry (1985) proved a positive density of primes $q$ with $P(q-1) > q^{2/3}$ (where $P(m)$ denotes the largest prime factor of $m$). This implies step 2 finds $r = O(\log^3 n)$, giving total cost $\widetilde{O}(\log^{15/2} n)$.

**To $\widetilde{O}(\log^6 n)$ (unconditional).** Lenstra and Pomerance (2003) found a provably $\widetilde{O}(\log^6 n)$ variant, without any conjecture.

**Conjectural improvement to $\widetilde{O}(\log^6 n)$.** The Sophie Germain Prime Density Conjecture (the number of Sophie Germain primes $q \le m$ is $\sim 2C_2 m/\ln^2 m$) would imply an $r = \widetilde{O}(\log^2 n)$ satisfying the step 2 condition, giving step 5 cost $\widetilde{O}(\log^6 n)$.

| **Result** | **Time** | **Status** |
|---|---|---|
| AKS original | $\widetilde{O}(\log^{21/2} n)$ | Unconditional |
| With Fouvry's theorem | $\widetilde{O}(\log^{15/2} n)$ | Unconditional |
| Lenstra–Pomerance variant | $\widetilde{O}(\log^6 n)$ | Unconditional |
| Under Sophie Germain conjecture | $\widetilde{O}(\log^6 n)$ | Conditional |
| Under Agrawal's conjecture | $\widetilde{O}(\log^3 n)$ | Likely false |

# 7 Open questions

**Reducing the loop count.** Step 5 runs for $\ell = \lfloor \sqrt{\varphi(r)} \log n \rfloor$ iterations. If a smaller generating set for $\hat{G}$ could be proved sufficient, $\ell$ would shrink and the exponent would improve.

**Agrawal's conjecture.** If $r$ is prime, $r \nmid n$, and $(X-1)^n \equiv X^n - 1 \pmod{X^r - 1, n}$, then either $n$ is prime or $n^2 \equiv 1 \pmod r$. This was verified computationally for $r \le 100$, $n \le 10^{10}$. If true, a prime $r \in [2, 4\log n]$ with $r \nmid n^2 - 1$ could be found (since the product of primes below $4\log n$ exceeds $n^2$), and checking the single congruence would give time $\widetilde{O}(\log^3 n)$. However, Lenstra and Pomerance (2003) gave a heuristic argument that the conjecture is likely *false*; variants with a larger lower bound on $r$ may still hold.

# 8 The complete proof in one place

Here is the entire argument distilled into ten logical steps:

1. **The identity (Lemma 2.1).** $(X+a)^n \equiv X^n + a \pmod n$ holds for all $a$ with $\gcd(a, n) = 1$ if and only if $n$ is prime. Proof: the middle binomial coefficients $\binom{n}{i}$ all vanish mod $n$ iff $n$ is prime.

2. **The compression.** Testing the identity exactly costs $\Omega(n)$ time. Working mod $X^r - 1$ compresses each polynomial to degree $< r$, costing $\widetilde{O}(r \log^2 n)$ per test.

3. **Choosing $r$ (Lemma 4.3).** We need $\mathrm{ord}_r(n) > \log^2 n$ to ensure later groups are large. Such $r \leq \log^5 n$ always exists.

4. **The loop.** Test $(*)$ for $a = 1, \ldots, \ell = \lfloor \sqrt{\varphi(r)} \log n \rfloor$. Any failure implies composite. All passing implies proceeding.

5. **Seeds of introspectivity.** Let $p \mid n$ be any prime. From (B) and (C), both $p$ and $n/p$ are introspective for each $(X + a)$.

6. **Closure.** Lemma 4.5: products of introspective numbers are introspective. Lemma 4.6: introspectivity is preserved under polynomial multiplication. So all of $I = \{(n/p)^i p^j\}$ is introspective for all of $P$ (products of the $(X + a)$'s).

7. **Group $G$.** Residues of $I$ mod $r$ form a subgroup of $\mathbb{Z}_r^*$ of size $t > \log^2 n$ (guaranteed by step 2's condition on $\mathrm{ord}_r(n)$).

8. **Group $\hat{G}$.** Images of $P$ in $\mathcal{F} = \mathbb{F}_p[X]/(h)$ form a subgroup of $\mathcal{F}^*$. Lower bound (Lemma 4.7): $|\hat{G}| > n^{\sqrt{t}}$, because distinct low-degree polynomials in $P$ map to distinct elements—otherwise their difference has $t$ roots but degree $< t$, impossible.

9. **Upper bound (Lemma 4.8).** If $n$ is not a power of $p$, then $\hat{I}$ has $> t$ elements but only $t$ residues mod $r$, so two agree (pigeonhole), forcing every element of $\hat{G}$ to satisfy a polynomial of degree $\leq n^{\sqrt{t}}$. So $|\hat{G}| \leq n^{\sqrt{t}}$.

10. **Conclusion.** The bounds $|\hat{G}| > n^{\sqrt{t}}$ and $|\hat{G}| \leq n^{\sqrt{t}}$ contradict each other. So $n = p^k$ for some $k$. Step 1 ruled out $k > 1$, so $n = p$ is prime.

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These notes are a self-contained companion to Agrawal, Kayal, Saxena, "PRIMES is in P," *Annals of Mathematics* **160**(2) (2004), 781–793.